

AI MANUFACTURING DECISION COPILOT

From Product Requirements to Evidence-Grounded Sourcing Decisions

Evidence Grounding • Constraint Checking • Transparent Supplier Ranking

Field	Detail
Hackathon	Sofistica Hackathon 2026
Challenge	AI Manufacturing Decision Copilot
Selected Track	Track 1 — Supplier Shortlisting
Team	S. Ali Raza, Huzaifa Ali
Date	August 2026
GitHub Repository	https://github.com/AliRaza2408/Copilot.git
Live Demo	https://drive.google.com/file/d/1ve3MHISHiTFpT_hr407MV1ienRXFmH8_/view?usp=sharing
Demo Video	https://drive.google.com/file/d/1ve3MHISHiTFpT_hr407MV1ienRXFmH8_/view?usp=sharing

1. Executive Summary

AI Manufacturing Decision Copilot is an AI-powered decision-support system designed to help product, sourcing, procurement, and manufacturing-operations teams make faster, more transparent, and evidence-grounded sourcing decisions.

Manufacturing decisions often require information from multiple documents, including product requirements, supplier profiles, quotations, technical specifications, certifications, quality records, lead-time information, and logistics assumptions. Reviewing and comparing this information manually can be time-consuming and can lead to missed requirements, inconsistent comparisons, or decisions that are difficult to trace back to their original evidence.

The Copilot allows users to upload relevant manufacturing and supplier documents. It processes these documents, extracts important information, organizes it into structured evidence, retrieves relevant evidence for user queries, evaluates suppliers against mandatory requirements, and produces a transparent recommendation.

The system is designed around an evidence-grounded architecture. Instead of simply asking an AI model to generate an answer, the Copilot connects important claims to their original source documents, and distinguishes between verified facts, assumptions, calculations, and AI-generated recommendations.

The system is a decision-support tool, not an autonomous procurement system. Supplier approval, external communication, quotation requests, and purchasing decisions remain under explicit human control.

2. Problem Statement

Manufacturing and procurement decisions often require information from many different sources. A typical manual process moves from product requirements through document review, supplier information gathering, spreadsheet comparison, and manual evaluation before a procurement decision is reached — a process that creates several challenges.

2.1 Information Is Distributed

Important information may exist across PDF documents, Excel spreadsheets, DOCX files, CSV files, technical specifications, supplier profiles, quotations, certification documents, and quality reports. Users must manually search through these documents to find relevant information.

2.2 Requirements Can Be Missed

A product may have mandatory requirements involving material, dimensions, weight, quantity, manufacturing capability, certification, quality, lead time, capacity, location, and sustainability. Manually checking every requirement against every supplier can result in missed constraints.

2.3 Supplier Information Is Inconsistent

Different suppliers express similar information in different formats — for example, one supplier states a single lead time, another splits it into production time and shipping time, and a third gives only an approximate figure. Although these values may represent similar information, they require normalization before comparison.

2.4 Missing Information Must Not Be Assumed

A supplier may not provide information about an important requirement. The system should not automatically assume the requirement is satisfied: evidence available leads to a PASS or FAIL result, while evidence missing must be reported as UNKNOWN.

2.5 Difficult-to-Trace Decisions

A simple AI system might simply state that one supplier is the best option. The user is then left asking why, which document supports that, and where a given value came from. This system addresses that problem by connecting every decision to its supporting evidence.

3. Proposed Solution

The proposed solution transforms manufacturing and supplier documents into structured, evidence-grounded decision support, helping users answer questions such as which supplier satisfies all mandatory requirements, why one supplier ranked higher than another, which requirements are unmet, what information is missing, what evidence supports a recommendation, and what happens if priorities change.

3.1 Selected Track — Supplier Shortlisting

The system identifies suppliers that satisfy mandatory product requirements and ranks eligible suppliers according to defined priorities, evaluating factors such as manufacturing capability, minimum order quantity, certification, location, capacity, quality history, lead time, sustainability requirements, and product specifications.

Crucially, the system performs an eligibility check before ranking. A supplier that fails even one mandatory requirement is excluded from ranking altogether — this prevents a supplier with a low price or fast lead time from ranking highly despite failing a mandatory requirement.

4. Key Innovations & Differentiation

The project is not intended to be simply "ChatGPT for procurement." Its main differentiation comes from combining the following:

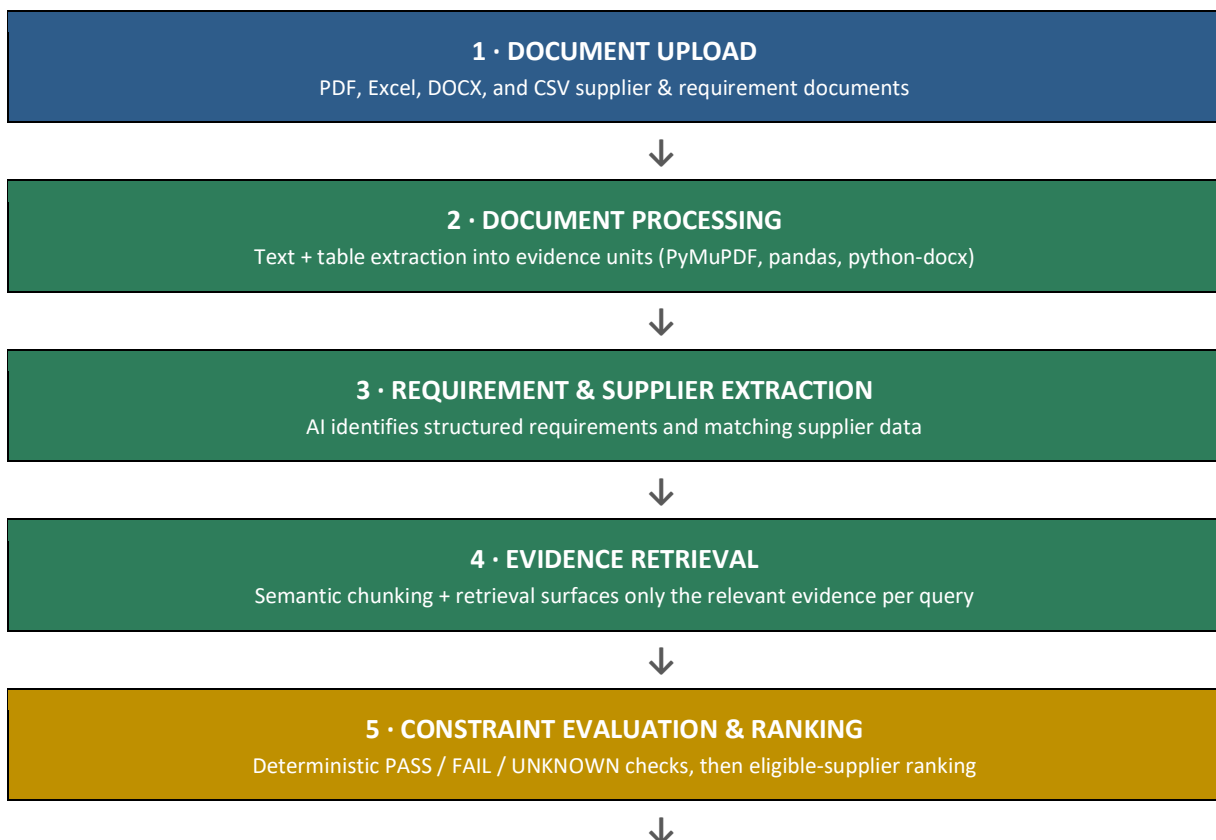
- Evidence grounding — important claims are connected directly to source evidence.
- Structured decision support — the system provides requirements, constraints, supplier comparisons, evidence, risks, and recommendations, rather than only conversational text.
- Multi-document reasoning — information distributed across several files is combined automatically.
- Requirement-aware evaluation — suppliers are evaluated against explicit, extracted requirements.
- Human-in-the-loop control — the AI assists the decision-maker instead of autonomously making procurement decisions.
- Failure-aware design — missing information becomes UNKNOWN, conflicting information becomes CONFLICT, and unavailable sources trigger fallback to human review, rather than inventing information.

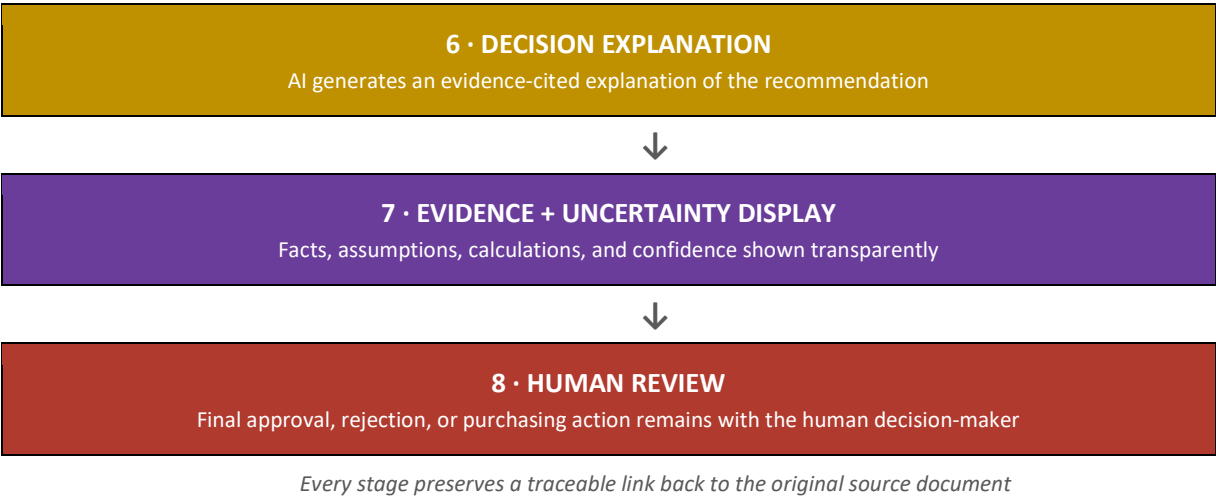
5. Technology Stack

Layer	Technology
Frontend	React
Backend	FastAPI
Programming	Python / JavaScript
PDF Processing	PyMuPDF
Excel Processing	pandas / openpyxl
DOCX Processing	python-docx
CSV Processing	pandas
AI / LLM	Llama
Vector Database	FAISS

6. System Workflow — Visual Flowchart

The diagram below shows how a single uploaded document flows through the evidence pipeline and becomes a transparent, traceable sourcing recommendation.





■ Upload ■ AI Processing ■ Decision Output ■ Transparency ■ Human Control

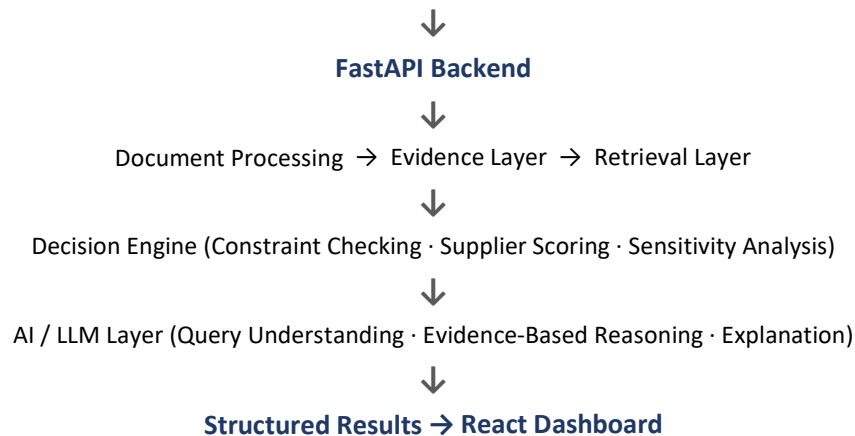
6.1 Step-by-Step Workflow

#	Stage	Description
1	Document Upload	User uploads product requirements, supplier profiles, quotations, and quality records.
2	Document Processing	Backend extracts text and tables into evidence units, preserving file name, page, and section metadata.
3	Document Normalization	Differently formatted values (e.g., '25 days' vs '3 weeks') are normalized without discarding the original source evidence.
4	Semantic Chunking	Documents are split into meaningful sections (e.g., MOQ, certification, lead time) rather than arbitrary page-sized chunks.
5	Requirement Extraction	AI identifies mandatory product requirements (material, weight, quantity, lead time, certification) as structured data.
6	Supplier Information Extraction	AI extracts the corresponding supplier data needed to evaluate each requirement.
7	Evidence Retrieval	User questions trigger retrieval of only the relevant evidence, with exact source and page cited.
8	Constraint Evaluation	Deterministic logic checks each mandatory requirement, returning PASS, FAIL, or UNKNOWN — never assuming a pass.

7. System Architecture

The technical architecture separates deterministic, programmatic logic from AI-driven language understanding, so that reliability does not depend on the language model alone.

React Frontend



8. Evidence-Grounded Decision Making

Instead of stating that "Supplier A is the best supplier," the system produces explanations such as: Supplier A is currently the highest-ranked eligible supplier because it satisfies the mandatory material, weight, lead-time, and certification requirements, with supporting evidence available in the supplier technical documentation. Users can inspect every underlying piece of evidence behind that statement.

8.1 Facts, Assumptions, Calculations, and Recommendations

- **Fact** — information directly supported by a source (e.g., a supplier reports an MOQ of 1,500 units).
- **Assumption** — an assumption introduced for analysis (e.g., the challenge-pack exchange-rate assumption applies to all quotations).
- **Calculation** — a deterministic result (e.g., a supplier MOQ of 1,500 against a required quantity of 2,000 yields a PASS).
- **Recommendation** — a decision-support conclusion (e.g., Supplier A is currently ranked first among eligible suppliers).

This separation improves transparency and reduces hallucination risk.

9. Limitations

- **Document quality** — poorly formatted or scanned documents may reduce extraction accuracy.
- **Complex tables** — some PDFs contain highly complex tables that require specialized extraction or OCR.
- **Missing information** — a supplier may not provide all required information.
- **Conflicting information** — different documents may contain different values for the same field.
- **AI limitations** — language models can still produce incorrect interpretations, which is why evidence grounding and human review are necessary.
- **Supplier verification** — information extracted from supplier documents should not automatically be treated as independently verified real-world truth.
- **External data** — the current system does not assume access to live supplier availability, real-time pricing, or real production capacity.

10. Safety and Responsible Use

The project follows a human-in-the-loop approach. The system must not contact real suppliers, send RFQs, approve suppliers, place purchase orders, or make binding procurement decisions without explicit human authorization.

The system also avoids presenting inferred information as verified fact — for example, stating that a supplier document reports a given capacity, rather than asserting that capacity as verified real-world availability.

10.1 Failure Handling

- Missing information — reported as UNKNOWN with a reason, and flagged for human verification, rather than assumed to be satisfied.
- Conflicting information — flagged as CONFLICT with the range of reported values, rather than silently selecting one.
- No relevant evidence — the system states that no supporting evidence was found, rather than hallucinating a value.

11. Testing and Validation

Test	Expected Result	Status
PDF Upload	File successfully processed	[TEST]
DOCX Upload	File successfully processed	[TEST]
Excel Upload	Data successfully extracted	[TEST]
CSV Upload	Data successfully extracted	[TEST]
Requirement Extraction	Requirements correctly identified	[TEST]
Supplier Extraction	Supplier data correctly identified	[TEST]
Evidence Retrieval	Relevant evidence retrieved	[TEST]
Constraint Checking	Correct PASS / FAIL / UNKNOWN	[TEST]
Supplier Ranking	Correct ranking	[TEST]
Missing Information	UNKNOWN generated	[TEST]
Conflicting Evidence	Conflict detected	[TEST]
Frontend → Backend	API communication works	[TEST]

12. Evaluation Metrics

- Mandatory Constraint Satisfaction Rate — how often the system correctly determines whether mandatory requirements are satisfied.
- Evidence Citation Coverage — what percentage of important claims contain supporting evidence.
- Citation Correctness — whether cited evidence actually supports the claim made.
- Hallucination / Unsupported Claim Rate — how often the system makes unsupported claims.

- Recommendation Agreement — how often the system agrees with the challenge-pack evaluation procedure.
- Cost Error and Lead-Time Error — accuracy against available reference values.
- Completion Time — time taken compared with the manual baseline.
- Human Review Effort — how much manual effort is required after using the Copilot.

13. Future Improvements

13.1 Advanced Document Intelligence

- Better OCR and table extraction
- Image/document understanding and handwritten document support

13.2 Advanced Retrieval & Supplier Intelligence

- Hybrid keyword + vector search with reranking and document version tracking
- Supplier reliability scoring, historical performance, and delivery-performance analysis

13.3 Cost & Risk Intelligence

- Quotation normalization, currency conversion, freight/duty calculation, and landed-cost comparison
- Supply disruption scenarios, alternative supplier recommendations, and lead-time risk analysis

13.4 Enterprise Integration

- Integration with ERP systems, procurement systems, supplier databases, and internal document repositories

14. Conclusion

AI Manufacturing Decision Copilot is designed to transform fragmented manufacturing and supplier information into a transparent, evidence-grounded sourcing decision.



Documents → Processing → Requirement Extraction → Evidence Retrieval → Constraint Evaluation → Ranking → Explanation → Human Decision

The most important principle of the system is that AI should not simply generate an answer. It should find the relevant information, preserve its source, extract structured facts, identify missing or conflicting information, apply explicit constraints, perform deterministic calculations where possible, rank eligible suppliers transparently, explain the recommendation using evidence, clearly communicate uncertainty, and keep final procurement decisions under human control.

This approach makes the Manufacturing Decision Copilot more suitable for real-world decision support than a general-purpose conversational AI system, because the focus is not only on generating an answer, but on producing a defensible, traceable, and reviewable manufacturing decision.